Solstice Task 1

February 15, 2026

§1 When you didn't just have an ex but also had y

Let $D_m = \{(x_i, y_i) \in (X^n, Y^n)\}$

1. **Posterior Probability:**
$$P(w|D_m) = \frac{\varphi(w) \times \prod_{i=1}^{m} P(y_i|x_i, w)}{\int \varphi(w) \prod_{i=1}^{m} P(y_i|x_i, w)\, dw}$$

2. **Marginal Likelihood:**
$$= \int \varphi(w) \prod P(y_i|x_i, w)\, dw$$

3. **Posterior Expectation:**
$$E[w] = \int w \cdot P(w|D_m)\, dw$$

   Posterior Variance:
$$Var[w] = E[w^2] - E[w]^2$$

   Note: $E[w^2] = \int w^2 P(w|D_m)\, dw$

4. **Predictive Function:**
$$P(y|x, D_m) = \int P(y_i|x_i, w) \times P(w|D_m)\, dw$$

§2 Gradient Descent (G.D) vs Bayesian approach

Loss Function:
$$L = \sum_{i=1}^{3} (y_i - (mx_i + c))^2$$

Dataset:

x	y
0	1
1	2
2	3

Expanded Loss:
$$L = (1 - c)^2 + (2 - (m + c))^2 + (3 - (2m + c))^2$$

Derivatives:
$$\frac{\partial L}{\partial m} = 2(2 - m - c)(-1) + 2(3 - 2m - c)(-2)$$

$$= -2(m + c - 2) - 4(2m + c - 3)$$

$$\boxed{\frac{\partial L}{\partial m} = 10m + 6c - 16}$$

$$\frac{\partial L}{\partial c} = -2(1 - c) + 2(2 - m - c)(-1) + 2(3 - 2m - c)(-1)$$

$$\boxed{\frac{\partial L}{\partial c} = -12 + 2c + 6m}$$

Iterations

1. INITIALISE: $m = 0, c = 0, \eta = 0.1$

2. Calculate Gradients:

$$\frac{\partial L}{\partial m} = -16, \quad \frac{\partial L}{\partial c} = -12$$

Update:

$$m \leftarrow 0 - 0.1(-16) = 1.6$$

$$c \leftarrow 0 - 0.1(-12) = 1.2$$

3. Next Step:

$$\frac{\partial L}{\partial m} \text{ using new values} \ldots$$

$$m_{new} = 1.6 - 0.1(7.2) = 0.88$$

$$c_{new} = 1.2 - 0.1(4.8) = 0.72$$

Convergence:

$$|m_{new} - m_{old}| < 10^{-5} \text{ or any other metric}$$

$$|c_{new} - c_{old}| < \ldots$$

Learning Rate (η):

- η small $\implies$ Very Slow Conv. Iter $\uparrow\uparrow$
- η Too large $\implies$ Overshooting

The Bayesian Approach

Prior:

$$P(m, c) \propto e^{-\frac{(m^2+c^2)}{2}} \quad \{\text{Normal Assumption}\}$$

Likelihood:

$$P(D|m, c) = \prod_i e^{-\frac{1}{2}(y_i - \hat{y}_i)^2} \quad \text{where } \hat{y}_i = mx_i + c$$

Model's assumption: Deviation b/w Data, Noise is shaped as Gaussian.

Posterior:

$$P(m, c|D) \propto P(m, c) \times P(D|m, c)$$

$$= e^{-\frac{1}{2}\left[(m^2+c^2)+(1-c)^2+(2-m-c)^2+(3-2m-c)^2\right]}$$

Calculation Strategy

No Step Wise Calculation $\longrightarrow$ **Surface Equation**

$$f(m,c) = m^2 + c^2 + (1-c)^2 + \dots \quad \text{is a Quadratic Bowl}$$

- # Min of this bowl $\Rightarrow$ Most Likely Sol^n
- # Width of bowl tells about Uncertainty.

[Diagram: A 2D grid representing m and c]

- Each cell has an associated Posterior Probability Value.

1. **MAP (Maximum A Posteriori):**

$$\hat{m}, \hat{c} = \underset{m,c}{\text{argmax}} \left[e^{-\frac{1}{2} f(m,c)} \right] = \underset{m,c}{\text{argmin}} \left[f(m,c) \right]$$

2. **Expected Value Method:**

$$\hat{m}_{avg} = \sum_{m,c} m \times P(m,c)$$

$$\hat{c}_{avg} = \sum_{m,c} c \times P(m,c)$$

3. **Updating the Prior:**
   - Initial Prior tells $(0,0)$ is Highest Probable initial chord.
   - How do we update the Prior?

$$\Rightarrow \text{Posterior}_i = \text{Prior}_{i+1}^{th} \text{ Iteration.}$$

Question: But then AGAIN aren't we doing G.D (Gradient Descent) type step leading to solution?

§3 Posterior Probability Function

$$P(w) = Ce^{-||w||^2} + 100e^{-10||w-w_0||^2}$$

$$P(w) = C \times \left[e^{-(\omega_1^2 + \omega_2^2 + \dots)} + 100 \times e^{-10((\omega_1-\omega_0)^2 + (\omega_2-\omega_0)^2 + \dots)} \right]$$

- If w was constraint $\Rightarrow$ Use Lagrange Multipliers.
- Else simply solve by $\frac{\partial P}{\partial \omega_i} = 0$ for $i = 1$ to N.

$$\frac{\partial P}{\partial \omega_i} = -2\omega_i e^{-(\omega_1^2+\omega_2^2+\dots)} - 2000(\omega_i - \omega_0)e^{-10(\omega_1-\omega_0)^2+\dots}$$

$$\Rightarrow \omega_i \left[e^{-(\sum \omega_j^2)} + 1000e^{-10(\sum(\omega_j-\omega_0)^2)}\right] = 1000\omega_0 e^{-10\sum(\omega_j-\omega_0)^2}$$

$$\boxed{\omega_i = \frac{1000\omega_0 e^{-10\sum(\omega_j-\omega_0)^2}}{1000e^{-10\sum(\omega_j-\omega_0)^2} + e^{-(\sum \omega_j^2)}}}$$

By Symmetry, $\omega_1 = \omega_2 = \cdots = \omega_m = \alpha$ (Say).

$$\alpha = \frac{1000e^{-100(\alpha-1)^2}}{1000e^{-100(\alpha-1)^2} + e^{-10\alpha^2}}$$

If assume $\alpha \to 1$:

$$\alpha = \frac{1000}{1000 + e^{-10}} \approx 1 \quad \text{(Hence proved 1)}$$

$$E[w] = \int \underbrace{wP(w) * K}_{\text{Posterior Expectation}} \quad dw$$

In Vector form:

$$E[\omega] = \begin{bmatrix} \int_{R^{10}} \omega_1 P(\omega)\, d\omega_1 d\omega_2 \dots d\omega_{10} \\ \int_{R^{10}} \omega_2 P(\omega) \dots \\ \vdots \\ \int_{R^{10}} \omega_{10} P(\omega)\, d\omega_1 d\omega_2 \dots d\omega_{10} \end{bmatrix} \times K$$

Standard Integral:

$$\int_R e^{-x^2} dx = \sqrt{\pi} \quad \Rightarrow \quad \int_{R^9} e^{-(\omega_2^2+\dots+\omega_{10}^2)} = (\sqrt{\pi})^9$$

Term-1 Calculation:

$$\Rightarrow \int_{R^{10}} \omega_1 P(\omega)\, d\omega_1 d\omega_2 \cdots = \int_R \omega_1 (\sqrt{\pi})^9 e^{-\omega_1^2} d\omega_1 + 100 \int_R \omega_1 e^{-10(\omega_1-\omega_0)^2} (\sqrt{\pi})^9 d\omega_1$$

Substitution: Put $\omega_1 - \omega_0 = t \implies d\omega_1 = dt$

$$\int_{-\infty}^{\infty} \omega_1 e^{-\omega_1^2} \dots \quad \text{[First term vanishes due to odd function if bounds are symmetric]}$$

Second part analysis:

$$\cdots + 100(\sqrt{\pi})^9 \int (t + \omega_0) e^{-10t^2} dt$$

$$= 100(\sqrt{\pi})^9 \omega_0 \int_{-\infty}^{\infty} e^{-10t^2} dt$$

(Since $\int te^{-10t^2} dt = 0$ as it is an odd function).

Solving the Gaussian integral: Let $-10t^2 = v \ldots$ or directly using standard form:

$$\int e^{-10t^2} dt = \sqrt{\frac{\pi}{10}}$$

$$\Rightarrow 100 \frac{\pi^{5/2}}{10^{5/2}\sqrt{10}} \int e^{-u^2}\, du = \boxed{\frac{100}{\sqrt{10}^5} \omega_0 \pi^5}$$

So

$$E[\omega] = \frac{100}{\sqrt{10}^5} \omega_0 \pi^5 \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \simeq \frac{\omega_0 \pi^5}{1000} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$= 0.306K \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \qquad [\because \omega_0 = 1]$$

$$\int \omega_1 p(\omega)\, d\omega_1 d\omega_2 \ldots d\omega_{10} = \frac{100}{\sqrt{10}} \pi^5 \cdot \omega_0$$

Normalization Const K:

$$\frac{1}{K} = Z = \int P(\omega)\, d\omega_1 d\omega_2 \ldots d\omega_{10}$$

$$= \left(\sqrt{\frac{\pi}{1}}\right)^{10} + 100 \left(\sqrt{\frac{\pi}{10}}\right)^{10} = \pi^5 + \frac{\pi^5}{1000}$$

$$\Rightarrow \text{Final Sol} = \frac{\left(\frac{\pi^5}{1000}\right)}{\left(1 + \frac{1}{1000}\right)\pi^5} = \boxed{\frac{1}{1001}}$$

Thus $E[\omega] = \frac{1}{1001} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \simeq 0$ Vector

Hence Proved 2

§4 Losses and Errors

4.1 Entropy

Surprisal: $:= -\log(p)$

- Rare Event $(p \downarrow) \Rightarrow$ High Surprisal
- Common Event $(p \uparrow) \Rightarrow$ Low Surprisal
- $[p = 1 \Rightarrow 0$ surprisal$]$

So surprisal is like a way to measure how surprising an event is.

(i) **Entropy is the Expected Value of Surprisal.**

$$\Rightarrow \text{Entropy is the AVG. level of SURPRISES/UNCERTAINTY in the distribution.}$$

$$\text{High Entropy} \longrightarrow \text{on Avg, very surprising} \Rightarrow \text{SPREAD OUT PDF}$$
$$\text{Low Entropy} \longrightarrow\Rightarrow \text{CONCENTRATED PDF}$$

(ii) Intuitively, the "Surprise" should satisfy *additivity* for 2 independent Events.

$$\Rightarrow \underline{\text{LOG}}$$

$$P(A \cap B) = P(A)P(B) \Rightarrow \text{Log can make them } \underline{\text{additive}}$$
$$\log(P(AB)) = \log(P(A)) + \log(P(B))$$

Proof of Equations:

The empirical entropy is given by:

$$S_n = -\frac{1}{n}\sum_{i=1}^{n} \log(q(x_i))$$

1. **Expectation of S_n:**

$$\mathbb{E}[S_n] = -\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[\log(q(x_i))]$$

Now,

$$\mathbb{E}[\log(q(x_i))] = \int \log(q(x_i))q(x_i)\,dx_i = -S$$

$$\Rightarrow \mathbb{E}[S_n] = -\frac{1}{n}\cdot n(-S) = \underline{\underline{S}}$$

2. **Variance of S_n:**

$$\text{Var}[S_n] = \mathbb{E}[S_n^2] - (\mathbb{E}[S_n])^2$$

First, calculating $\mathbb{E}[S_n^2]$:

$$\begin{aligned}
\mathbb{E}[S_n^2] &= \frac{1}{n^2}\mathbb{E}\left[\left(\sum_{i=1}^{n}\log(q(x_i))\right)^2\right] \\
&= \frac{1}{n^2}\mathbb{E}\left[\sum_{i=1}^{n}\log^2(q(x_i)) + 2\sum_{1\le i<j\le n}\log(q(x_i))\log(q(x_j))\right] \\
&= \frac{1}{n^2}\left\{\sum_{i=1}^{n}\mathbb{E}[\log^2(q(x_i))] + 2\sum_{1\le i<j\le n}\mathbb{E}[\log(q(x_i))\log(q(x_j))]\right\}
\end{aligned}$$

Using linearity and the fact that independent variables allow expectation splitting (x_i, x_j independent):

$$= \frac{1}{n^2}\left[n\int \log^2(q(x))q(x)dx + 2\sum_{1\le i<j\le n}\underbrace{\mathbb{E}[\log(q(x_i))]}_{-S}\cdot\underbrace{\mathbb{E}[\log(q(x_j))]}_{-S}\right]$$

The number of terms in the summation $\sum_{1\le i<j\le n}$ is $\frac{n(n-1)}{2}$.

$$= \frac{1}{n^2}\left[n\int q(x)\log^2(q(x))dx + 2\cdot\frac{n(n-1)}{2}\cdot(-S)(-S)\right]$$

$$= \frac{1}{n^2}\left[n\int q(x)\log^2(q(x))dx + n(n-1)S^2\right]$$

Thus,

$$\mathbb{E}[S_n^2] = \frac{1}{n}\int q(x)\log^2(q(x))dx + \frac{n-1}{n}S^2$$

Now substituting back into the Variance formula:

$$\begin{aligned}
\mathrm{Var}[S_n] &= \left[\frac{1}{n}\int q(x)\log^2(q(x))dx + \frac{n-1}{n}S^2\right] - S^2 \\
&= \frac{1}{n}\int q(x)\log^2(q(x))dx + S^2\left(\frac{n-1}{n} - 1\right) \\
&= \frac{1}{n}\int q(x)\log^2(q(x))dx + S^2\left(\frac{n-1-n}{n}\right) \\
&= \frac{1}{n}\int q(x)\log^2(q(x))dx - \frac{1}{n}S^2
\end{aligned}$$

Thus:

$$\mathbb{V}[S_n] = \frac{1}{n}\left[\int q(x)\log^2(q(x))dx - S^2\right]$$

Hence Proved

4.2 Generalization and Training Loss

:

$$G_m = -\int q(x)\log\left[p(x|x^n)\right]dx$$

$$S = -\int q(x)\log(q(x))dx$$

$$G_m - S = \int q(x)\log\left(\frac{q(x)}{p(x|x^n)}\right)dx$$

- $G_m - S$ is the Kullback-Leibler (KL) Divergence between $q(x)$ and predictive $p(x|x^n)$.

$$G_m - S = D_{KL}\left(q(x)||p(x|x^n)\right)$$

- Gibbs Inequality $\rightarrow G_m \geq S$
  Thus, $G_m = S$ iff $q(x) = p(x|x^n)$
  ie, only if predictive model is EXACTLY the real distribution.

1. Can Generalization loss be calculated in real world?
   No. As $q(x)$ is needed [and we dont have that].
2. Can CV loss be calculated?
   YES, as no $q(x)$ is needed in its expression.

but so is Training loss. T_m/C_m both dont have $q(x)$. So then WHY DONT WE USE T_m ?

Ans: Training loss vs Cross Validation Loss

- $T_m = \frac{-1}{m}\sum_{i=1}^{m}\log(p(x_i|x^n))$
- Here we would end up testing on X_i when we already trained on X_i
- <u>Train on $X_1, X_2 \dots X_m$ Test on X_1</u>
  ...
  Then Avg
- $\Rightarrow$ Model may memorize X^n
- [G_m will be Large]
- T_m Tells How well you the model memorized the past

- [Leave one-out Cross Validation [LOOCV]]
- We neglect 1 Point at a time to maximize amount of data model gets to study
- We CAN neglect k more $\Rightarrow$ [k fold]
- $C_m = \frac{-1}{m}\sum_{i=1}^{m}\log(p(x_i|x^n \setminus x_i))$
- Hide $X_1 \rightarrow$ Train on $X_{2\dots} \rightarrow$ Test X_1
- " " $X_2 \rightarrow$ " " $X_1, X_3 \cdots \rightarrow$ Test X_2
- Then Avg all scores
- C_m tells How well u predict future
- $\Rightarrow$ # It is an Estimator of G_m

<u>So</u>: G_m is Ideal, C_m is good, T_m is bad.
(but not calculatable)

3. CV depends on?
   - The dataset X^n and the predictive model chosen.
   - It uses available data to approximate unseen test error.
4. <u>Bonus</u>: # Proof that $E[C_m] = E[G_{m-1}]$:

$$E[C_m] = \frac{1}{m}\sum E\left[\log P(x_i|x^n \setminus x_i)\right]$$

Let $X^n \setminus X_i$ be [Training set] be written as D_{m-1}
Then $E[C_m] = \frac{1}{m}\sum E\left[\log(P(x_{test}|D_{m-1}))\right]$
Now, $E\left[\log p(x_{test}|D_{m-1})\right] = -E[G_{m-1}]$

$$\Rightarrow E[C_m] = \frac{1}{m}\sum_{i=1}^{m}(-E[G_{m-1}])$$

$$\boxed{E[C_m] = -E[G_{m-1}]}$$

<u>Proof of</u>

$$E[\log(p(x_{test}|D_{m-1}))] = -E[G_{m-1}]$$

We know that $G_m = -E_{gen}[\log(p(x_{test}|D))]$ (overall test data)

$$-E[G_m] = E[\log(p(x_{test}|D))]$$

($\uparrow$ overall train data)

Comparing Variances of $G_m - S$ and $C_m - S_m$:

<u>PODASIP</u> $\Rightarrow$ Disprove

- T_m is bad metric compared to G_m as it is biased (Trained on text set as well)
- In Complex model $\longrightarrow$ It can memorize X^n (Overfitting)
- Overfitting $\Rightarrow T_m$ Very Low [Close to S_m as $T_m \geq S_m$]
  G_m is Very high.

- So comparing models based on T_m is incorrect.

Salvage: G_m is better than T_m [Theoretical] as it can measure KL divergence from S. Practically C_m is better as G_m not Calculatable.

4.3 CROSS-VALIDATION - LOSS

:

- Cross Validation Error $= C_m - S_m$
- Suppose we have a chaotic dataset. Now no matter what curve we make, the feasibility of the model is going to be bad as we cant cover the random peaks and valleys. This is bad because even if our model is close by, it will give a bad score. so we try to normalise it by subtracting the entropy. This makes comparing various datasets more fair. Similarly, if the dataset is relatively smooth, our answer must give a solution that is close by and shouldnt get extra help from the entropy function which is exactly what is happening

PODASIP :

1. True. (Already proved in 4-2)
2. **FALSE** :
   We define the predictive density for a data point x_i given the dataset excluding that point, denoted as $x^n \setminus x_i$, as:

$$p(x_i \mid x^n \setminus x_i) = \int p(x_i \mid w) \cdot p(w \mid x^n \setminus x_i)\, dw \tag{1}$$

   The Cross-Validation (CV) loss is defined as the average negative log predictive density:

$$\text{CV loss} = -\frac{1}{n} \sum_{i=1}^{n} \log \left(\int p(x_i \mid w) \cdot p(w \mid x^n \setminus x_i)\, dw \right) \tag{2}$$

   We need to find the relationship between the full posterior $p(w \mid x^n)$ and the leave-one-out posterior $p(w \mid x^n \setminus x_i)$.
   By Bayes' theorem, the leave-one-out posterior is:

$$p(w \mid x^n \setminus x_i) = \frac{1}{Z_{-i}} \cdot p(x^n \setminus x_i \mid w) \cdot p(w) \tag{3}$$

   where $p(w)$ is the prior and Z_{-i} is the normalizing constant.
   We know that the full likelihood is the product of the leave-one-out likelihood and the single point likelihood:

$$p(x^n \mid w) = p(x^n \setminus x_i \mid w) \cdot p(x_i \mid w)$$

   Therefore:

$$p(x^n \setminus x_i \mid w) = \frac{p(x^n \mid w)}{p(x_i \mid w)}$$

   Substituting this into the proportionality relation:

$$\begin{aligned} p(w \mid x^n \setminus x_i) &\propto \frac{p(x^n \mid w) \cdot p(w)}{p(x_i \mid w)} \\ &\propto \frac{p(w \mid x^n)}{p(x_i \mid w)} \end{aligned}$$

   To find the exact distribution, we must normalize. The normalizing constant is the integral of the numerator:

$$Z = \int \frac{p(w \mid x^n)}{p(x_i \mid w)}\, dw = \mathbb{E}_{w \sim p(w \mid x^n)} \left[\frac{1}{p(x_i \mid w)} \right]$$

Thus, the closed-form expression for the leave-one-out posterior is:

$$p(w \mid x^n \setminus x_i) = \frac{\frac{p(w|x^n)}{p(x_i|w)}}{\int \frac{p(w|x^n)}{p(x_i|w)}\, dw} \tag{4}$$

If we substitute equation (5) back into equation (1), we get a simplified expression for the predictive density that does not require re-training the model n times:

$$\begin{aligned} p(x_i \mid x^n \setminus x_i) &= \int p(x_i \mid w) \cdot \frac{\frac{p(w|x^n)}{p(x_i|w)}}{Z}\, dw \\ &= \frac{1}{Z} \int p(w \mid x^n)\, dw \\ &= \frac{1}{Z} \cdot 1 \\ &= \frac{1}{\int \frac{p(w|x^n)}{p(x_i|w)}\, dw} \end{aligned}$$

Therefore, the leave-one-out predictive density is the harmonic mean of the likelihood evaluated at samples from the full posterior:

$$p(x_i \mid x^n \setminus x_i) = \frac{1}{\mathbb{E}_{w \sim p(w|x^n)} \left[\frac{1}{p(x_i|w)} \right]} \tag{5}$$

Substituting the result for the leave-one-out predictive density back into the definition of the cross-validation loss (equation 2):

$$\begin{aligned} C_n &= -\frac{1}{n} \sum_{i=1}^{n} \log p(x_i \mid x^n \setminus x_i) \\ &= -\frac{1}{n} \sum_{i=1}^{n} \log \left(\frac{1}{\mathbb{E}_{w \sim p(w|x^n)} \left[\frac{1}{p(x_i|w)} \right]} \right) \end{aligned}$$

Using the property $\log(1/a) = -\log(a)$, the negative signs cancel out:

$$\begin{aligned} C_n &= -\frac{1}{n} \sum_{i=1}^{n} \left(-\log \left(\mathbb{E}_{w \sim p(w|x^n)} \left[\frac{1}{p(x_i \mid w)} \right] \right) \right) \\ &= \frac{1}{n} \sum_{i=1}^{n} \log \left(\mathbb{E}_{w \sim p(w|x^n)} \left[\frac{1}{p(x_i \mid w)} \right] \right) \end{aligned}$$

Thus, the final derived expression for the Cross-Validation loss is:

$$C_n = \frac{1}{n} \sum_{i=1}^{n} \log \left(\mathbb{E}_{w \sim p(w|x^n)} \left[\frac{1}{p(x_i \mid w)} \right] \right) \tag{6}$$

Thus we proved it false and salvaged it with the correct expression

3. Cauchy Schwarz: $E[Y^2] \geq E[Y]^2$ [or that $\mathrm{Var}[Y] \geq 0$]

First, we express both losses in terms of the expectation over the posterior $p(w|X^n)$. Let $k(w) = p(X_i|w)$ denote the likelihood of a single data point.

From our previous derivation, the Cross-Validation Loss (using the importance sampling approximation) is the log of the expected inverse likelihood (harmonic mean):

$$
\begin{aligned}
C_n &= -\frac{1}{n}\sum_{i=1}^{n}\log\left(\frac{1}{\mathbb{E}_{w|X^n}[1/k(w)]}\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}\log\left(\mathbb{E}_{w|X^n}\left[\frac{1}{k(w)}\right]\right) \qquad (7)
\end{aligned}
$$

Applying Jensen's Inequality The function $f(x) = \frac{1}{x}$ is strictly convex for $x > 0$. By Jensen's Inequality ($\mathbb{E}[f(x)] \geq f(\mathbb{E}[x])$), we have:

$$
\mathbb{E}_{w|X^n}\left[\frac{1}{k(w)}\right] \geq \frac{1}{\mathbb{E}_{w|X^n}[k(w)]}
$$

Taking the logarithm of both sides (since log is monotonically increasing):

$$
\log\left(\mathbb{E}_{w|X^n}\left[\frac{1}{k(w)}\right]\right) \geq \log\left(\frac{1}{\mathbb{E}_{w|X^n}[k(w)]}\right)
$$

Using the property $\log(1/a) = -\log(a)$, the right side becomes:

$$
\log\left(\mathbb{E}_{w|X^n}\left[\frac{1}{k(w)}\right]\right) \geq -\log\left(\mathbb{E}_{w|X^n}[k(w)]\right)
$$

We sum this inequality over all n data points and divide by n:

$$
\frac{1}{n}\sum_{i=1}^{n}\log\left(\mathbb{E}_{w|X^n}\left[\frac{1}{k(w)}\right]\right) \geq -\frac{1}{n}\sum_{i=1}^{n}\log\left(\mathbb{E}_{w|X^n}[k(w)]\right)
$$

Substituting the definitions from (1) and (2):

$$
C_n \geq T_n
$$

hence proved

□

§5 Free Energy

Proof of $\int Z(x^n)\,dx_1dx_2\ldots dx_m = 1$:

$$
\Rightarrow \int\left(\int \varphi(\omega)\prod_{i=1}^{n} p(x_i|\omega)\,d\omega\right)dx_1dx_2\ldots dx_m
$$

$$
\Rightarrow \int \varphi(\omega)\left[\int\prod_{i=1}^{m} p(x_i|\omega)\,dx_1dx_2\ldots dx_m\right]d\omega
$$

$$
\Rightarrow \int \varphi(\omega)\times\prod_{i=1}^{m}\left[\int p(x_i|\omega)\,dx_i\right]d\omega
$$

$$
= \int \varphi(\omega)d\omega \quad \left[\because \int p(x_i|\omega)dx_i = 1\right]
$$

$$
= 1
$$

Thus:

$$\boxed{\int Z(x^n)\, dx_1 dx_2 \ldots dx_m = 1}$$

$$E[F_m] - mS = E\left[-\log(Z(x^n))\right] + \int q(x^n)\log(q(x^n))\, dx$$
$$= \int -\log(Z(x^n))q(x^n)\, dx + \int q(x^n)\log(q(x^n))\, dx$$

$$E[F_n] - nS = \int q(x^n)\log\left(\frac{q(x^n)}{Z(x^n)}\right)\, dx^n$$

Which is KL divergence between $q(x^n)$ and $Z(x^n)$

$$\text{Thus, } \boxed{E[F_n] - nS = D_{KL}\left(q(x^n)||Z(x^n)\right)}$$

Note: Entropy is additive $\Rightarrow$ So as $S = \int q(x)\log(q(x))\, dx$,

$$\Rightarrow mS = \int q(x^n)\log(q(x^n))\, dx$$

[$\because$ Log converts Products to addition]

$$G_m - S = +\int q(x)\log\left(\frac{q(x)}{p(x|x^n)}\right)\, dx = D_{KL}\left(q(x)||p(x|x^n)\right)$$

$$\Rightarrow \boxed{E[G_n] - S = E\left[D_{KL}(q(x)||p(x|x^n))\right]}$$

So: $E[G_n] - S$ is Generalization error $\Rightarrow KL$ b/w True and Predictive $(q(x)||p(x|x^n))$
$E[F_n] - nS$ is KL b/w True and Marginal Likelihood $(q(x^n)||Z(x^n))$

Reflection:

We can establish a direct relationship between the cumulative quantity $\mathbb{E}[F_n] - nS$ and the instantaneous quantity $\mathbb{E}[G_n] - S$.

The marginal likelihood of the sequence $Z(X^n)$ can be factorized into conditional predictive densities:

$$Z(X^n) = \prod_{i=1}^{n} p(X_i \mid X^{i-1}) \tag{8}$$

Taking the negative logarithm of both sides gives the Free Energy F_n:

$$\begin{aligned} F_n &= -\log Z(X^n) \\ &= -\sum_{i=1}^{n} \log p(X_i \mid X^{i-1}) \\ &= \sum_{i=1}^{n} G_{i-1} \end{aligned}$$

where G_{i-1} is the Generalization Loss at step $i-1$.

Now, subtracting the total entropy nS (which is $\sum S$):

$$\begin{aligned}\mathbb{E}[F_n] - nS &= \mathbb{E}\left[\sum_{i=1}^{n} G_{i-1}\right] - \sum_{i=1}^{n} S \\ &= \sum_{i=1}^{n} (\mathbb{E}[G_{i-1}] - S)\end{aligned}$$

Interpretation: This equation reveals that the **Cumulative Prediction Error** (KL divergence of the full sequence) is the sum of the **Expected Generalization Errors** (KL divergences of the individual predictive distributions) over all time steps $k = 0$ to $n - 1$:

$$D_{KL}(q(X^n) \parallel Z(X^n)) = \sum_{k=0}^{n-1} \mathbb{E}\left[D_{KL}(q(x) \parallel p(x \mid X^k))\right]$$

- $\mathbb{E}[G_n] - S$ represents the **instantaneous rate of learning** (how much error the model makes on the *next* prediction).
- $\mathbb{E}[F_n] - nS$ represents the **total cost of learning** incurred over the entire history of the process.

§6 Summing Up

$p(x|a) = a^x(1-a)^{1-x}$ $\boxed{\text{a is the } \omega \text{ here}}$

1. **Likelihood** $= \prod_{i=1} P(x_i|a) = a^{x_1+x_2+\cdots} \times (1-a)^{m-(x_1+x_2+\cdots)} = L(\text{say})$

$$\begin{aligned}\log(L) &= (\sum x)\log(a) + (m - \sum x)\log(1-a) \\ &= (m_1 \log a) + m_2 \log(1-a)\end{aligned}$$

Differentiating WRT a:

$$\frac{L'}{L} = \frac{m_1}{a} - \frac{m_2}{1-a}$$

$L' = 0 \Rightarrow a = \hat{a} \Rightarrow \frac{m_1}{\hat{a}} = \frac{m_2}{1-\hat{a}}$

$$\Rightarrow m_1 - m_1\hat{a} = m_2\hat{a} \Rightarrow \hat{a} = \frac{m_1}{m_1+m_2} = \frac{m_1}{m}$$

Thus, $\boxed{\hat{a} = \frac{m_1}{m}}$

2. $p(x|x^n) = \int p(x|a)p(a|x^n)\,da$

$$= \int a^x(1-a)^{1-x} \times \frac{\varphi(a) \times \text{Likelihood}}{\int \varphi(a) \times \text{Likelihood}\,da}\,da$$

$\varphi(a) = 1$, Likelihood $= L = a^{m_1}(1-a)^{m_2}$

Also $\int \varphi(a)L\,da = \int a^{m_1}(1-a)^{m_2}\,da$

$$\Rightarrow p(x|x^n) = \frac{\int a^x(1-a)^{1-x} \times a^{m_1}(1-a)^{m_2}\,da}{\int a^{m_1}(1-a)^{m_2}\,da}$$

$$\Rightarrow p(x|x^n) = \frac{B(x+m_1+1, 2-x+m_2)}{B(m_1+1, m_2+1)}$$

$$\Rightarrow p(1|x^n) = \frac{B(2+m_1, 1+m_2)}{B(m_1+1, m_2+1)} = \frac{\Gamma(m_1+2)\Gamma(m_2+1)}{\Gamma(m_1+1)\Gamma(m_2+1)} \cdot \frac{\Gamma(m_1+m_2+2)}{\Gamma(m_1+m_2+3)}$$
$$= \frac{(m_1+1)\times 1}{m_1+m_2+2} = \boxed{\frac{m_1+1}{m+2}}$$

$$p(0|x^n) = \frac{B(m_1+1, 2+m_2)}{B(m_1+1, m_2+1)} = \frac{(m_2+1)\times 1}{m_1+m_2+2} = \boxed{\frac{m_2+1}{m+2}}$$

<u>So</u>: DISPROVE, SALVAGE $\rightarrow p(0|x^n) = \frac{m_2+1}{m+2}$

3. We aim to derive the generalization error defined by the Kullback-Leibler (KL) divergence from the true distribution $p(x|a_0)$ to the estimated distribution $p(x|\hat{a})$.

For a Bernoulli distribution where $p(x=1|a) = a$ and $p(x=0|a) = 1-a$:

$$\begin{aligned} D_{KL}(p(x|a_0) \parallel p(x|\hat{a})) &= \sum_{x\in\{0,1\}} p(x|a_0) \log \frac{p(x|a_0)}{p(x|\hat{a})} \\ &= p(1|a_0) \log \frac{p(1|a_0)}{p(1|\hat{a})} + p(0|a_0) \log \frac{p(0|a_0)}{p(0|\hat{a})} \\ &= a_0 \log \frac{a_0}{\hat{a}} + (1-a_0) \log \frac{1-a_0}{1-\hat{a}} \end{aligned}$$

Expanding the logarithmic terms:

$$K = a_0 \log a_0 - a_0 \log \hat{a} + (1-a_0)\log(1-a_0) - (1-a_0)\log(1-\hat{a}) \tag{9}$$

To avoid singularities when $n_1 = 0$ or $n_2 = 0$ (which would cause $\log(0) \rightarrow -\infty$), we use a smoothed estimator with a small positive value ϵ:

$$\hat{a}_{ML} = \frac{n_1+\epsilon}{n+2\epsilon} \tag{10}$$

Consequently:

$$1-\hat{a}_{ML} = 1 - \frac{n_1+\epsilon}{n+2\epsilon} = \frac{n+2\epsilon-n_1-\epsilon}{n+2\epsilon} = \frac{n_2+\epsilon}{n+2\epsilon} \tag{11}$$

where $n_2 = n - n_1$.

Substituting $\hat{a}_{ML}$ and $1-\hat{a}_{ML}$ into Equation (9):

$$\begin{aligned} K_{ML} = a_0 \log a_0 &+ (1-a_0)\log(1-a_0) \\ &- a_0 \log\left(\frac{n_1+\epsilon}{n+2\epsilon}\right) - (1-a_0)\log\left(\frac{n_2+\epsilon}{n+2\epsilon}\right) \end{aligned}$$

This matches the expression provided in the problem statement.

From Part 2, the Bayesian predictive distribution with a uniform prior is

$$p(1|X^n) = \hat{a}_{Bayes} = \frac{n_1+1}{n+2} \tag{12}$$

Similarly:

$$p(0|X^n) = 1-\hat{a}_{Bayes} = \frac{n_2+1}{n+2} \tag{13}$$

Substituting these values into the general KL divergence formula:

$$
\begin{aligned}
K_{Bayesian} &= a_0 \log a_0 + (1-a_0)\log(1-a_0) \\
&\quad - a_0 \log(\hat{a}_{Bayes}) - (1-a_0)\log(1-\hat{a}_{Bayes}) \\
&= a_0 \log a_0 + (1-a_0)\log(1-a_0) \\
&\quad - a_0 \log\left(\frac{n_1+1}{n+2}\right) - (1-a_0)\log\left(\frac{n_2+1}{n+2}\right)
\end{aligned}
$$

Comparing the two results, we observe that the Bayesian generalization error is equivalent to the smoothed Maximum Likelihood error when $\epsilon = 1$.

$$
K_{Bayesian} \equiv K_{ML}\Big|_{\epsilon=1} \tag{14}
$$